Topic 21 — Species & Ecology (पारिस्थितिकी)

Covers syllabus (click to expand)

Red Data Book (रेड डेटा बुक) | IUCN (आईयूसीएन) Red List Categories | Wetlands | Mangroves | Coral Reefs | Grasslands | Himalayan Ecosystem (पारिस्थितिकी तंत्र) | Marine Ecosystem

Sources baked in: NCERT Bio 12 Ch 15–16; Geo 11; IUCN Red List; Ramsar 1971; BSI/ZSI; MoEFCC; UPPCS PYQs 2018–2025 **Weight:** ★★★ Very High — IUCN threatened trap; UP Ramsar geography; coral bleaching; Himalayan zones **Last verified:** August 2026

Current Affairs (this topic)

- New IUCN category moves for flagship species (प्रतीक (फ्लैगशिप) प्रजाति) are the usual CA refresh.
- Keystone / umbrella / flagship / indicator definitions stay concept-fixed.

21.1 Red Data Book

Identity: National endangered-species catalogue | BSI plants | ZSI animals | Kolkata HQ | ≠ IUCN Red List

- A **Red Data Book** lists **endangered, rare, and threatened species (संकटग्रस्त प्रजातियां)** of a country or region — "red" signals conservation urgency.
- In **India**, **BSI (Botanical Survey of India)** publishes the **Red Data Book of Indian Plants**.
- **ZSI (Zoological Survey of India)** publishes the **Red Data Book of Indian Animals**.
- Both **BSI and ZSI** are headquartered in **Kolkata**.
- IUCN pioneered the early Red Data Book concept in the **1960s**, which evolved into the modern **IUCN Red List** with standard global categories.
- **Red Data Book ≠ IUCN Red List:** Red Data Book is **national documentation**; IUCN Red List is a **global standardised assessment** (EX through NE).
- **Red Data Book ≠ CITES:** Red Data Book documents status; **CITES** regulates **international trade** in listed species.
- Species may appear in Red Data Books, **IUCN Red List**, and **WPA Schedules** simultaneously — three parallel systems (documentation, global assessment, legal protection).

Feature	Red Data Book (India)	IUCN Red List
Scope	National	Global
Publisher	BSI (plants), ZSI (animals)	IUCN (Gland)
Function	National threat inventory	Standardised risk categories

> **Logic:** "Red Data Book of Indian plants published by ZSI" = **FALSE** — plants = **BSI**, animals = **ZSI**.

21.2 IUCN Red List Categories

Identity: 9 categories | Threatened = VU+EN+CR | IUCN Gland | Assesses risk ≠ CITES trade

- The **IUCN Red List** assigns assessed species to **nine categories** based on population (जनसंख्या), decline rate, range, and extinction probability.
- **Extinct (EX):** No reasonable doubt the last individual has died.
- **Extinct in the Wild (EW):** Survives only in **captivity, cultivation, or seed banks** — no viable wild population.
- **Critically Endangered (CR) (अति संकटग्रस्त):** Extremely high extinction risk — e.g., **gharial**, **Great Indian Bustard**.
- **Endangered (EN) (लुप्तप्राय):** Very high risk — e.g., **Gangetic dolphin**, **tiger** (improved from CR in some assessments).
- **Vulnerable (VU):** High risk but lower urgency than EN/CR — e.g., **sloth bear**.
- **Near Threatened (NT):** Close to qualifying as threatened but not yet.
- **Least Concern (LC):** Widespread, low extinction risk.
- **Data Deficient (DD):** Insufficient data — not the same as safe.
- **Not Evaluated (NE):** Not yet assessed — majority of world's species.
- **Threatened = VU + EN + CR** — the most tested formula. **Endangered alone is only one-third of threatened.**
- **Risk hierarchy** among living categories: **CR > EN > VU > NT > LC**.
- **IUCN assesses extinction risk.CITES** regulates **trade** — separate bodies, separate functions.

Code	Category
EX	Extinct
EW	Extinct in Wild
CR	Critically Endangered
EN	Endangered
VU	Vulnerable
NT	Near Threatened
LC	Least Concern
DD	Data Deficient
NE	Not Evaluated

> **Logic:** "All threatened species are Endangered" = **FALSE**. Threatened = **VU + EN + CR**.

21.3 Wetlands

Identity: Ramsar 1971 | Wise Use | 2 February | UP Ramsar geography trap

- **Wetlands** are ecosystems where **water saturates the soil** — marshes, swamps, bogs, floodplains, lake margins — permanent or seasonal.
- They are among the **most productive ecosystems**, providing flood control, groundwater recharge, water purification, fisheries, and migratory bird habitat (वास स्थान).
- The **Ramsar Convention (1971)** (रामसर कन्वेंशन) promotes **conservation and wise use** of wetlands. **World Wetlands Day = 2 February**.
- A **Ramsar Site** is a wetland of **international importance** nominated by the contracting country — separate from WPA national park status.
- **India** has **75+ Ramsar sites** (2024 milestone). **Amrit Dharohar Scheme (2023)** supports wetland conservation and livelihoods.
- **UP Ramsar sites:Sarsai Nawar** (Etawah (इटावा)), **Samaspur Bird Sanctuary** (Raebareli), **Nawabganj** (Unnao), **Upper Ganga River**, **Bakhira**, **Haiderpur**, **Parvati Agra Bird Sanctuary**.
- **NOT in UP:Rudrasagar Lake** = **Tripura**. **Sultanpur National Park** = **Gurugram, Haryana**. Both are Ramsar sites outside UP.
- **Threats:** drainage for agriculture, pollution (eutrophication (सुपोषण)), invasive species (आक्रामक विदेशी प्रजाति) (**water hyacinth** chokes UP ponds), hydrological alteration.

PYQ — UPPCS Prelims 2025, Q58

Which Ramsar sites are NOT located in Uttar Pradesh?

1. Sarsai Nawar Jheel 2. Samaspur Bird Sanctuary 3. Rudrasagar Lake 4. Sultanpur National Park

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

Show answer

Ans: B (3 and 4) — Rudrasagar = **Tripura**; Sultanpur = **Haryana**. 1 and 2 are in UP.

Type	India example
Freshwater marsh	Keoladeo (केवलादेव) Ghana (घन)
Riverine wetland	Upper Ganga (UP)
High-altitude wetland	Chandra Taal, Tso Moriri

Type	India example
Coastal/mangrove wetland	Sundarbans

> **Logic:** Ramsar = **wetlands**, not climate or forests.

21.4 Mangroves

Identity: Halophytes | Coastal tidal | Pneumatophores | Sundarbans | MISHTI 2023-24

- **Mangroves** are **salt-tolerant** trees and shrubs growing in **intertidal mudflats** of tropical and subtropical coasts.
- **Pneumatophores** (vertical aerial breathing roots) and **prop roots** (Rhizophora) enable gas exchange in waterlogged anaerobic soil.
- **Viviparous germination:** seeds germinate **while attached to the parent tree** and drop as ready seedlings (propagules).
- **Ecological roles:** coastal protection against cyclones/tsunamis, fish nursery grounds, sediment trapping, shoreline stability, **blue carbon** (नीला कार्बन) sequestration.
- Mangroves act as a **safety hedge** because trees **do not get uprooted easily** — they have **extensive, deep root systems** (not because of a wide uninhabited buffer alone).
- **Coastal Odisha** is India's most cyclone-prone Bay of Bengal coast; large mangrove loss in the **Mahanadi (महानदी) delta (डेल्टा)** increases cyclone risk (classic A/R pair).
- **Sundarbans** (West Bengal + Bangladesh) is the **largest mangrove forest** — **Sundari** tree is the flagship associate; royal Bengal tiger and estuarine crocodile.
- **Bhitarkanika** (Kendrapara, Odisha) sits on the **Brahmani–Baitarni** delta — **not** Vansadhara–Subarnarekha, and **not** West Bengal. Ramsar site (**2002**).
- **Coringa** mangrove (Andhra Pradesh) lies in the **Godavari** (गोदावरी) delta — do not confuse with Coondapur/Honnavar (Karnataka coast names).
- Other Indian mangrove areas: **Pichavaram** (TN), **Gulf of Kutch / Khambhat** (Gujarat), **Andaman** (अंडमान) coasts, **Goa** (गोवा) (e.g. Chorao protected mangrove).
- Mangrove area rank (ISFR 2021 style): **West Bengal > Gujarat > Andaman & Nicobar (~616–617 km²) > Andhra Pradesh**.
- **Andaman & Nicobar** uniquely combine **mangrove + evergreen + deciduous** forests in one geography.
- **UP is landlocked** — no natural mangrove coast; questions test national ecology and schemes.
- **MISHTI** (Mangrove Initiative for Shoreline Habitats and Tangible Income) was announced in **Union Budget 2023–24** to restore mangrove forests and support livelihoods.

PYQ — UPPCS Prelims 2025, Q93

MISHTI: (1) It was initiated in 2019. (2) It aims to restore mangrove forests.

A. Only 2

B. Neither

C. Both

D. Only 1

Show answer

Ans: A — Restoration aim is correct. MISHTI = **2023–24 Budget**, not 2019.

Species	Feature
Rhizophora	Stilt/prop roots
Avicennia	Salt-secreting leaves
Sonneratia	Pneumatophores

> **Logic:** Mangroves = **coastal saline**, not freshwater marshes alone.

21.5 Coral Reefs

Identity: $CaCO_3$ polyps + zooxanthellae | Warm shallow seas | Bleaching = heat | India four (चातुर्याम) reef regions

- **Coral reefs** are underwater **calcium carbonate structures** built by **coral polyps** (Cnidaria) in warm, shallow, clear seas.
- **Zooxanthellae** (symbiotic dinoflagellate algae) live inside coral tissue and supply up to **90% of coral energy** via photosynthesis.
- **Coral bleaching:** under stress (especially **high temperature**), corals **expel zooxanthellae** and turn white. Prolonged stress causes mortality.
- Growth needs warm water (**18–30°C**), shallow depth (**<30 m** for light), low turbidity, normal salinity.
- **Reef types:fringing** (attached to coast), **barrier** (lagoon between reef and shore), **atoll** (ring around submerged volcano).
- **India's coral reef areas:Gulf of Mannar** (first marine biosphere reserve (जैवमंडल आरक्षित क्षेत्र)), **Lakshadweep** (लक्षद्वीप), **Andaman & Nicobar**, **Gulf of Kutch** (northernmost Indian reefs).
- **Threats:** ocean warming (primary bleaching driver), acidification, sedimentation, destructive fishing, pollution.

PYQ — UPPCS Prelims 2024, Q19

Coral bleaching in Andaman & Nicobar caused by:

A. Increased industrial activity

B. Increase of 2°C in Indian Ocean

C. Heavy marine pollution

D. Decrease of 4°C in Indian Ocean

Show answer

Ans: B — Heat stress from **+2°C warming** bleaches coral. Cooling does not cause bleaching.

Logic: Bleaching = **temperature rise**, not pollution alone or **–4°C cooling**.

21.6 Grasslands

Identity: Grass-dominated | Fire + grazing | Terai UP | Dudhwa barasingha | Kaziranga rhino

- **Grasslands** are ecosystems **dominated by grasses** with few trees — maintained by **grazing, fire, and climate**.
- **Tropical savanna (सवाना):** warm climate, wet-dry seasons, scattered trees — Indian **semi-arid Deccan (दक्कन)** and **Vidarbha**.
- **Terai grasslands (UP):** Himalayan foothills with **tall elephant grass** (Saccharum) and sal forest mosaic. **Dudhwa NP** protects tiger, rhino, **barasingha (swamp deer)**.
- **Floodplain grassland:Kaziranga** (Assam) — tall grass supports one-horned rhino.
- **Alpine meadow:** above **~3,500 m treeline** in Himalaya — short summer herb growth.
- **Shola-grassland mosaic (Western Ghats (पश्चिमी घाट)):** montane grass patches alternating with shola forest — Nilgiri (नीलगिरि) tahr habitat.
- **Threats:** conversion to cropland, overgrazing, invasive woody species (lantana), fire suppression.

Type	Location	Key species
Terai grassland	UP foothills	Barasingha, tiger — Dudhwa
Floodplain grassland	Assam	Rhino — Kaziranga
Semi-arid savanna	Deccan	Blackbuck
Alpine meadow	High Himalaya	Medicinal herbs

Logic: Terai = UP grassland-forest mosaic**. Puszta (पुस्ज़टा) =** Hungary** (world geography distractor), not Indian grassland.

21.7 Himalayan Ecosystem

Identity: Altitude → climate → vegetation belts | Snow leopard | Trans-Himalaya cold desert | Dudhwa Terai fringe

- The **Himalayan ecosystem** spans from tropical foothills to **nival (permanent snow) zone** — biodiversity changes with **altitude and climate**.

- **Altitudinal vegetation belts** (general sequence upward): **Tropical** (<1,000 m — sal, shisham) → **Subtropical** (chir pine) → **Temperate** (deodar, fir) → **Sub-alpine** (rhododendron) → **Alpine meadow** → **Nival** (>4,500 m — lichens, snow).
- Temperature drops roughly **6.5°C per 1,000 m** (lapse rate), driving vegetation zonation.
- **Flagship fauna (प्राणीजात):** Terai/foothills — elephant, tiger; mid-Himalaya — musk deer, monal; high Himalaya — **snow leopard**, blue sheep.
- **Trans-Himalaya** (Ladakh (लद्दाख)/Spiti rain-shadow): cold desert, sparse vegetation — radically different from southern wet slopes.
- Himalayas are the **source of perennial rivers** (Ganga, Yamuna (यमुना), Brahmaputra (ब्रह्मपुत्र) tributaries) — snow and glacier melt feeds base flow.
- **UP link:** Terai and Shivalik (शिवालिक) foothills (Dudhwa) form the transition between Gangetic plain and Himalaya.

PYQ — UPPCS Prelims 2025, Q21

Assertion (A): In the Himalayan mountains different types of vegetation are found. Reason (R): In Himalayas, there are variations in climate with change in altitude.

A. Both true; R does not explain A

B. A false, R true

C. A true, R false

D. Both true; R explains A

Show answer

Ans: D — Altitude-driven climate change (जलवायु परिवर्तन) explains vegetation zonation — standard NCERT chain.

Zone	Altitude (approx.)	Vegetation
Tropical foothills	<1,000 m	Sal, riverine forest
Temperate	2,000–3,000 m	Deodar, fir
Alpine meadow	3,600–4,500 m	Herbs, dwarf shrubs
Nival	>4,500 m	Lichens, permanent snow

> **Logic:** Snow leopard = **high Himalayan** zones, not Terai or Sundarbans.

21.8 Marine Ecosystem

Identity: ~71% Earth surface | Neritic = productive | India 7516 km coast | Gangetic dolphin ≠ marine

- **Marine ecosystems** cover **~71% of Earth's surface** — estuaries, continental shelves, coral reefs, open ocean, deep sea.

- **Neritic zone** (continental shelf, **0–200 m**) has the **highest fisheries productivity** — light, nutrients, upwelling.
- **Pelagic zone** = open ocean water; **benthic zone** = seafloor organisms.
- **Euphotic zone** (0–200 m) allows photosynthesis; **aphotic** zone is dark.
- **Upwelling** (Arabian Sea west coast during SW monsoon (दक्षिण-पश्चिम)) brings nutrient-rich deep water — supports sardines, mackerel.
- **Estuaries** (river meets sea) are highly productive nurseries but pollution-sensitive.
- **India:~7,516 km coastline**, **Arabian Sea** (west) and **Bay of Bengal** (east), EEZ ~2.02 million km².
- **Gangetic dolphin** is a **freshwater river** mammal — **NOT marine** — common trap.
- **Threats:** plastic pollution, oil spills, overfishing, bottom trawling, warming/acidification affecting reefs.

Zone	Character	Productivity
Intertidal	Tide-exposed shore	Moderate (नरम दल)
Neritic	Continental shelf	**High**
Pelagic	Open ocean	Low–moderate
Benthic	Seafloor	Variable

> **Logic:** Highest marine fisheries productivity = **neritic/continental shelf**, not open deep ocean.